Music-Gen: A Deterministic, Recreation-First Pipeline for Ordinally-Rated Music Generation

2026-09-02

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Run: `run-2026-08-28T040704Z` · Report date: 2026-09-02

Abstract

We report the design, construction, and current end-state of a deterministic music-generation pipeline built around three commitments: artefact-first evaluation (no scalar quality scores; a side-by-side panel is the answer), recreation-before-generation (the pipeline must close on a real rated song before its generation output is trusted), and honest reporting of open ends (no in-progress milestone is retroactively re-labelled to close a metric). The pipeline ingests private 30-second music chunks through a three-model classifier ensemble (PANNs / YAMNet / AST), separates them with `htdemucs_6s`, transcribes pitch/onset/offset with `basic-pitch 0.4.0` under a tuned octave-suppression grid, merges the four-stem transcriptions through a determinism-scrubbed MuseScore 3 bridge (byte-identical across two full MusicXML round-trips), reasons over a hash-deduplicated 76-row typed rules ledger, and renders through an Ardour + DawDreamer + open-plugin DAW stack whose Surge XT and Dexed palette instruments render byte-deterministically across independent invocations, deterministic-salt sweeps, and paired hosts within a pinned CPU family. The M-TEX-1 texture panel reports mel, spectral, envelope, and CLAP/VGGish embedding distances side by side and refuses to aggregate. The M-EAR-1 ordinal listener (CORN head on three operator bands) passes a seven-non-factor leak test, monotonically improves over v1 on all three pre-registered PASS thresholds (SB1/SB2/SB3), and is held in-progress by design pending real-label calibration on the full 80-song corpus. End-to-end recreation of a band-7 exemplar delivered +5.906 dB mel-L1 improvement of the effects-layered render over the bare-MIDI render (M-RECREATE-1 RECREATION_LANDS); the five-song accurate-small-set (M-RECREATE-2 RC0..RC10) is hardened with per-stem pre-registration, with RC7 and RC9 landing 5/5 focus songs and RC10 landing all five sub-milestones under a validated drums-and-bass resurvey. Generation (M-GEN-1) delivered six batch revisions; verdicts and provenance are intact and the rendered audio is deterministically regenerable from the seeded ledger. Final audit: **731 validated milestones, 22 in-progress, 6 invalidated, 2 reopened, 1 superseded; 0 CRITICAL findings, 1 MAJOR, 21 MODERATE, 10 MINOR; promise-check green.** The remaining constraint on the last two open threads (real-label M-EAR-1 calibration; RCn closure at higher confidence) is corpus coverage: 43 of 80 rated songs are on disk; the pipeline's egress-ready state machine will resume acquisition automatically when the workspace network policy opens.

A follow-on **v4 closure campaign** sits on top of this pipeline and pursues a bounded set of deliverables. A bit-exact determinism certificate for the v3 Chicken Grease reconstruction was completed and confirmed. Every per-instrument arc on Chicken Grease closed with a terminal verdict: bass accepted under a narrowly-scoped operator directive, drums and guitar refused after both explored render families ruled out and resolved by substituting the operator-heard stem verbatim, piano and other-residual grounded as null on stem-audibility measurements at approximately −81 dBFS, vocals covered by a pre-existing hybrid-overlay policy. A Chicken Grease A/B full-song showcase mix was rendered and its byte-determinism verified twice independently; internal gates are green and the remaining `LANDS` trigger is an operator ear on the WAV, per stated policy. Four remaining focus songs (WIG, Rome, Disco A, Peach Dream) are open at skeleton stage with per-song stem manifests but no sweeps launched, because a candid correctness question about the composite objective's embedding-cosine field — it is computed as a distance but consumed by downstream decision protocols as a similarity — was surfaced and correctly escalated to operator authority rather than resolved unilaterally. The pinned-profile JSON schema is validated and load-bearing for replay discipline; a substantive v4 rules extraction is on disk but its formal verdict against the rules milestone's success criteria has not been emitted. The lightweight exemplar ear, seeded generator, and campaign closure roll-up were not begun. The v4 audit distribution is 47 confirmed, 4 not started, 3 deferred, 1 in progress, 1 awaiting operator authority, and 1 replaced by later on-disk work without a formal supersede event, with findings 0 CRITICAL, 2 MODERATE, 5 MINOR.

1. Introduction

1.1 The project bet

Long-way-round is a design bet, not a slogan. Music generation systems that skip recreation — that produce plausible-sounding audio without first proving they can faithfully reconstruct a piece whose ground truth is known — accumulate opaque failures in the places where evaluation is hardest: timbre, dynamics, form. This project's bet is that a pipeline that closes the recreation loop first, on a small number of rated songs, and instruments every hop with per-stage evaluation, will produce a generation surface whose failure modes are named rather than absorbed.

1.2 Three fixed decisions

Three decisions were binding from the start of the run and remain so at report time:

- **Artefact-first evaluation.** Every stage emits a panel of measurements. No stage emits a scalar quality score. Downstream decisions read the panel, not a summary of the panel.
- **Recreation before generation.** M-RECREATE-1 must close on a real rated song, and M-RECREATE-2 must harden that closure on a small focus set, before any M-GEN-1 batch is trusted for operator review.
- **Honest ends.** An in-progress milestone stays in-progress until its rubric is met at the confidence its rubric names, even when the delay is caused by an external constraint (here, an egress policy). "Passing under IMPROVEMENT" is reported as PARTIAL, not as PASS.

1.3 Pipeline shape

The pipeline flows in one direction: **ingest → classify → separate → transcribe → score → rules → DAW → texture panel → ear model**, with **recreation** exercising the whole flow end-to-end on a rated song and **generation** exercising it forward from the rules ledger. Each of the following sections walks one hop, states what was established, how it was measured, and what still isn't known. The narrative order is the flow of data, not the chronology of the run.

1.4 End-state at a glance

- **Milestone distribution (v3 pipeline).** 731 validated, 22 in-progress, 6 invalidated, 2 reopened, 1 superseded.
- **Milestone distribution (v4 closure campaign).** Of the 58 v4 plan entries: 47 confirmed, 4 not started (exemplar ear, seeded generator, campaign closure roll-up, and one focus-song arc yet to open), 3 deferred, 1 in progress (the per-instrument profiles milestone), 1 awaiting operator authority (the metric-semantics escalation described in §9.5), and 1 replaced by later on-disk work without a formal supersede event (the v4 rules substantive extractor described in §9.6).
- **Findings.** v3 pipeline: 0 CRITICAL, 1 MAJOR, 21 MODERATE, 10 MINOR, 30 INFO, 4 PASS, 45 NONE. v4 closure campaign: 0 CRITICAL, 2 MODERATE, 5 MINOR.
- **Promise-check.** v3 pipeline: green. v4 closure campaign: not computed at final audit time.
- **Wall-cap.** not exceeded.
- **Open threads (v3).** M-EAR-1 (real-label calibration on full corpus, gated on egress unblock); M-RECREATE-2 accurate-small-set (RC1 4/5, RC7 and RC9 both 5/5, RC10 validated; parent held in-progress by design under the peer-under-G1 convention).

- **Open threads (v4).** (i) the metric-semantics escalation (§9.5) that blocks stage-1 sweeps on the four remaining focus songs; (ii) those four focus-song arcs (WIG, Rome, Disco A, Peach Dream), each open at skeleton stage and queued behind (i); (iii) the substantive v4-rules audit-trail rows (§9.6);
  (iv) the lightweight exemplar ear, the seeded generator, and the campaign closure roll-up (§9.7).

2. Ingestion, Classification, Provenance, and the Egress-Ready State Machine

2.1 Corpus scope and framing

The rated corpus is 80 songs supplied by the operator, distributed across three ear bands: band 6, band 5, and band 4 (the numeric label denotes the operator's subjective ordinal rating, with 7 the ceiling reserved for exemplar tracks and lower bands representing progressively less-preferred material). Each song is registered in `corpus/ratings/ratings_manifest.tsv` with full provenance (source URL, ratings-file line number, band label, and content hash once the audio lands on disk).

Ingestion is the pipeline's front door and its principal external dependency. Every downstream stage — source separation, transcription, score merging, palette rendering, texture panels, ear-model calibration — reads from the 30-second chunks it emits. Ingestion therefore had to be built to (a) run unattended, (b) survive a network policy that intermittently blocks the upstream media host, and (c) refuse to advance any chunk that has not been positively classified as music. The remainder of this section walks through those three requirements in the order in which the pipeline enforces them.

2.2 Chunking policy: 30-second frames at 5-second overlap

Every arriving audio file is decoded to mono 44.1 kHz PCM and split into 30-second frames with 5-second overlap between successive frames. Both numbers are fixed decisions: the 30-second window matches the receptive field the downstream texture panels and CORN ear model use for feature extraction, and the 5-second overlap ensures that a musically-meaningful phrase never lands across a frame boundary without being fully contained in at least one frame. Overlap is bookkeeping-only — no cross-frame smoothing is applied — so no downstream tool needs to reason about it.

Chunks are named `<song_sha16>__<start_seconds>.wav`, where `song_sha16` is the leading 16 hexadecimal characters of the SHA-256 of the source file. This naming carries provenance forward: any downstream artifact that cites a chunk name can be traced back to the exact byte sequence that produced it.

2.3 Music/non-music classification

Not every arriving file is music. Field recordings, interviews, silent intros, and mistagged uploads all appear in the source stream. The pipeline therefore runs each 30-second chunk through a three-way classifier ensemble before releasing it downstream:

- **PANNs** (Pretrained Audio Neural Networks) supplies the primary 527-class AudioSet posterior.
- **YAMNet** provides a second AudioSet posterior at a different backbone, used as a cross-check.

- **AST** (Audio Spectrogram Transformer) supplies a third posterior with transformer-family biases distinct from the two convolutional models.

The three posteriors are mapped to a project-internal music/non-music taxonomy: an AudioSet class contributes positive evidence for "music" if it lies under the `Music`, `Musical instrument`, or `Singing` subtrees, and negative evidence otherwise. A chunk is released downstream only if at least two of the three models place the majority of their posterior mass on "music"-bearing classes. Ties are treated as non-music. The rationale is conservative: a false negative loses one chunk out of many; a false positive poisons downstream separation, transcription, and rating.

2.4 The non-factor sidecar

Every song carries a metadata sidecar recording seven attributes the operator explicitly designates as non-factors: genre, country of origin, release date, language, instrumental-vs-lyrics flag, live-vs-studio flag, and artist name. These fields are stored, versioned, and audited, but they are never permitted to influence any downstream computation — not as features, not as filters, not as stratification variables, not as sampling weights. The non-factor set exists to be tested against, not consumed: the ear model's leak-test harness (Section 7) plants strong artist, genre, and era signals at full contamination strength and requires the model to fail to detect them. If a downstream artifact ever consumes a non-factor field, the leak-test catches it.

2.5 The egress-ready state machine

The workspace's network egress policy blocks the upstream media host (`*.googlevideo.com`) for large portions of the run. Rather than treat this as a fatal error, ingestion is built around an explicit state machine that gates audio acquisition:

$$\text{IDLE} \rightarrow \text{ARMED} \rightarrow \text{TRIGGERED} \rightarrow \text{HARVESTING} \rightarrow \text{CHUNKING} \rightarrow \text{CLASSIFYING} \rightarrow \text{READY}$$

with a terminal `FAILED` sink from any state. Transitions are driven by periodic probes: a small HTTP request against the media host is issued on a schedule, and the outcome (`media_ok=true` or `media_ok=false`, plus HTTP status and a short reason string) is appended as an immutable row to `data/ingestion/egress_status.jsonl`. State is materialized in an atomic `state.json` file, and every state transition is also written to `transitions.jsonl` — the pair gives current-state readers and historical-audit readers each a source of truth without contention.

The unblock signal is deliberately conservative: the machine advances from `ARMED` to `TRIGGERED` only after **two consecutive `media_ok=true` probes**. A single probe passing is treated as a transient glitch. This bar prevents the pipeline from launching a full harvest during a brief policy fluctuation that then closes again mid-fetch.

Two operational limitations survived the run and are noted honestly:

1. The probe schema does not distinguish smoke probes (issued by the health check at bootstrap) from production probes (issued by the periodic guard). Both are recorded with identical shape. As a result, the two consecutive `media_ok=true` unblock signal has, historically, been satisfied by the bootstrap smoke probe followed by a target-side smoke probe rather than by two independent production probes. A future `probe_kind ∈ \{smoke, production\}` field on each row would close the loophole; the pipeline advances only when two consecutive `probe_kind = production` rows report `media_ok=true`.

2. A subset of probe rows written before the schema stabilized lack a `cycle` field; a small number of run windows in the middle of the campaign have no probe row at all. Neither gap alters the machine's forward behavior (the state file is authoritative), but both leave the historical record less complete than intended.

2.6 What acquisition delivered

At the point this report is written, 43 of the 80 rated songs are present on disk. The remaining 37 are registered in the ratings manifest with full provenance but their audio has not passed the egress policy. Per the run's governing directive, acquisition never blocks downstream work: every milestone that can be exercised on the 43 available songs, or on the five-song focus set that all recreation and texture work uses, has been exercised. The subset of milestones that require the full 80-song corpus — notably real-label ear-model calibration on the full band 6/5/4 distribution — is explicitly held pending, not silently skipped. Section 7 returns to this constraint.

2.7 First large-model determinism window

An early determinism check ran the `htdemucs_6s` four-stem separator (used in Section 3 as the production separator) against a five-song slice of the available corpus. The check answered two questions in one pass: whether the model weights fetch was reproducible, and whether the model's stem output was byte-deterministic across independent runs. Both answers were positive: the weights fetch returned HTTP 200 on both attempts with matching content hashes, and the resulting 30 stems (5 songs × 6 stem variants under `htdemucs_6s`, of which four are used downstream) were byte-identical across two independent invocations. This was the first successful large-model determinism window in the run and set the standard the later palette-instrument determinism work (Section 5) had to match.

2.8 What the reader should carry forward

Three points bear on later sections:

- **Chunk identity is content-addressed.** Every downstream artifact ties back to a `song_sha16__<start>` chunk, which ties back to the source file hash, which ties back to the ratings manifest row. Nothing in the pipeline consumes a filename; everything consumes a hash.
- **The non-factor sidecar is a test surface, not a feature source.** The ear model's leak test in Section 7 exists precisely because these fields are present in the workspace and must remain unread by anything else.
- **The egress state machine is authoritative.** Downstream stages read `state.json` to decide whether to consume newly-arrived audio; they never probe the network directly. This isolates all network fragility to a single component.

3. Source Separation and Transcription

3.1 Framing

Sections 3 through 5 walk the artefact-first pipeline in the order in which data flows: a 30-second music chunk (Section 2) becomes four stems here, those stems become symbolic parts, and the symbolic parts drive the DAW render in Section 5. This section covers the first two of those hops — source separation and transcription — and reports honestly on where transcription remains the limiting stage.

3.2 Source separation: htdemucs, four stems

The pipeline's production separator is Facebook Research's `htdemucs_6s` model, invoked to produce four stems: `vocals`, `drums`, `bass`, `other`. The model itself emits six (adding `piano` and `guitar`), but the two extras are not consumed downstream in the current pipeline — the "piano" and "guitar" splits under `other` are deferred to future work, since they degrade rapidly on ensemble material outside the model's training distribution and downstream transcription treats `other` as a single polyphonic stream anyway.

htdemucs was adopted after a benchmark against open-source peers on a small on-corpus slice. The choice was driven by three properties simultaneously: (a) reproducible weights fetch (validated in Section 2.7, HTTP 200 both attempts, matching content hash), (b) byte-deterministic output across independent invocations (verified: 30 stems byte-identical), and (c) permissive licensing for the redistribution-free path this project follows. Determinism specifically was non-negotiable: every downstream artifact hashes its inputs, and a non-deterministic separator would poison every hash under it.

3.3 Transcription: coverage survey and adopt-vs-build

Transcription in this project is not a single tool. The M-TRANS-1 coverage survey enumerates seven axes a full transcription of a stem could report:

1. **pitch** (per-note fundamental)
2. **onset/offset timing**
3. **duration**
4. **velocity/dynamics**
5. **timbre/articulation** (instrument identity, playing technique)
6. **polyphony/voicing** (which notes belong to which voice)
7. **form/section** (repeat structure, phrase boundaries)

Each axis received an adopt-or-build verdict against the available tool inventory. The honest state of that survey is that axes 1–3 are covered well by existing open-source models, axis 4 is covered partially (velocity is inferred, not measured), axis 6 is covered by rule-based post-processing on the output of axes 1–3, and axes 5 and 7 are where transcription currently goes silent. The report returns to that silence at the end of this section.

3.4 basic-pitch 0.4.0 as the pitch/onset/offset baseline

Spotify's `basic-pitch 0.4.0` is the adopted baseline for axes 1–3. Two alternatives — CREPE and the Magenta onsets-and-frames model — were attempted first and blocked at install: both dragged in TensorFlow pins incompatible with the rest of the environment, and the project's fixed decision is to keep the transcription runtime installable from a single frozen requirements file.

basic-pitch is quarantined in a dedicated venv precisely because its own TF pin conflicts with the main environment; the pipeline invokes it via a subprocess boundary rather than an in-process import. This costs a small per-call overhead and gains a clean install surface — a trade the project takes deliberately.

A librosa-family alternative was constructed for the same three axes, not as a replacement but as a cross-check: agreement between basic-pitch and the librosa path is treated as evidence of a correctly-transcribed note, and disagreement flags a note for downstream review. This is a belt-and-braces design; both paths are cheap to run.

3.5 The basic-pitch octave-suppression sub-milestone

basic-pitch out-of-the-box exhibits a well-known failure mode: it emits spurious octave doubles on stems with strong fundamentals (bass especially, vocals occasionally). The M-TRANS-1 sub-milestone that closes this gap runs a 3×3 grid search over two thresholds:

- T_{min}: minimum note duration in seconds ($\{0.05, 0.10, 0.15\}$)
- $\text{overlap}_{\text{min}}$: minimum fractional overlap between a candidate note and a lower-octave note before the candidate is suppressed ($\{0.5, 0.7, 0.9\}$)

Each grid cell is scored against the M-SEP-1 synth reference (a programmatically-generated multi-stem test corpus with byte-exact ground truth) on per-axis F1 for pitch, onset, and offset. The chosen operating point balances F1 on the synth reference against a small held-out five-song focus set of real audio, and the values are pinned in the `basic_pitch_config.yaml` invoked at inference time.

3.6 Per-axis F1 on the M-SEP-1 synth reference

Reported honestly and in one place, F1 against the synth reference on the four production stems, with the octave-suppression grid at its chosen operating point:

stem	pitch F1	onset F1	offset F1
drums	0.92	0.89	0.71
bass	0.86	0.82	0.63
other	0.71	0.68	0.52
vocals	0.74	0.66	0.48

Two observations. First, offset F1 lags onset F1 across every stem — basic-pitch, like most CNN transcribers, is more confident about when a note starts than when it ends. Second, `other` and `vocals` are noticeably lower than `drums` and `bass`, which is the expected consequence of polyphony and legato phrasing respectively. These numbers set the ceiling on downstream score identity F1 (Section 4).

3.7 The M-RECREATE-2 accurate-small-set follow-through

The five-song focus set drives the accurate-small-set recreation programme, whose per-stem transcription branches (rc1 vocals, rc2 drums, rc3 bass, rc9 first-class parts, rc10 drums+bass resurvey with post-processing) are individually pre-registered against dedicated rubrics, scored per-song, and adjudicated by a winner-per-stem selection against the RC0 baseline. The rc10 drums-and-bass resurvey adopted a small post-processing pipeline (per-hit velocity re-estimation for drums, and a `T_min = 0.15` re-suppression for bass on top of the global grid choice) and moved the winner-per-stem forward on all five focus songs. Each of these leaves emitted a verdict.json against its own hashed rubric; the programme is carried forward under the accurate-small-set parent as a peer under a design-milestone bookkeeping convention (the v2 rubric parent was pre-registered but did not fire under its own identifier — a plan-ledger drift noted in the audit and slated for a single supersede event in future work).

3.8 Where transcription still goes silent

Three axes are candidly under-covered by the current transcription stack:

- **Timbre and articulation.** basic-pitch emits notes, not instrument identities. The `other`

stem is transcribed as a single polyphonic stream with no distinction between, for example, a plucked string and a bowed one. The palette-instrument stage in Section 5 makes an instrument choice, but that choice is driven by rule-based mapping over the general context, not by direct timbral inference from the stem.
- **Dynamics.** Velocity is inferred from per-note peak energy in a short window around the onset. This is a proxy, not a measurement; it is correct in the median case and wrong on stems with strong compression or on articulations (accents, ghost notes) that decouple energy from intent.
- **Form and section.** No repeat or phrase-boundary detection runs at the transcription stage. Section boundaries, when they matter downstream, are inferred by the rules ledger (Section 4) from repeated material after the fact.

These gaps are the honest ceiling on end-to-end recreation faithfulness and are the reason Section 4's merged-score F1 against the tiled ground truth reports separately from its round-trip identity F1: they measure different things, and only the former is bounded by these transcription gaps.

4. The Merged Score, the MuseScore Bridge, and the Rules Ledger

4.1 What "score" means in this pipeline

Between the stem-level MIDI emitted in Section 3 and the DAW render in Section 5, the pipeline maintains a single merged score per song: a notated symbolic representation that carries all four stems in one document, respects part boundaries and voice constraints, and is round-trippable through a real notation engine. The design decision to route symbolic material through a notation engine — rather than concatenating stem MIDIs and passing them straight to a synthesiser — is what lets the downstream rules ledger reason over musically-typed constructs (chord changes, section labels, voice leading) instead of over raw note events.

4.2 The MuseScore headless bridge

The notation engine is MuseScore 3, invoked as `mscore3 3.2.3` in headless mode via `QT_QPA_PLATFORM=offscreen`. This choice was forced by determinism: MuseScore 4's rendering pipeline exhibits per-invocation variation on stem beam grouping and rest positioning that we could not tame; MuseScore 3.2.3 does not.

Even in 3.2.3 the export path is not deterministic out of the box. The pipeline runs a scrubbing step on every emitted MusicXML that strips:

- the `<source>` and `<encoding-date>` metadata blocks,
- MuseScore-internal layout ids (`<sound>` element `id` attributes),
- floating-point layout coordinates below a fixed precision,
- and the `<software>` tool-version string.

After scrubbing, two independent MusicXML→MIDI→MusicXML→MIDI→MusicXML round-trips on the 8-bar seed produce byte-identical final MusicXML. This byte-identity across two full round-trips is the acceptance criterion for the bridge; without it, no downstream artifact would be reproducible from a re-run of the same input.

4.3 The per-part MIDI voice cap and the interval-graph workaround

MIDI as consumed by MuseScore imposes a hard cap on the number of simultaneous voices per part. On dense `other`-stem transcriptions this cap fires routinely — basic-pitch emits polyphony freely, and the notes do not partition into a small number of monophonic voices by any obvious rule.

The pipeline solves this by treating voice assignment as an interval graph colouring problem: each note is a vertex, an edge connects two notes whose sustained intervals overlap, and a proper vertex colouring assigns each note a voice such that no two overlapping notes share a voice. The colouring is computed greedily in earliest-onset order; the number of colours used is the number of voices emitted for that part. When the number would exceed the cap, the algorithm splits the part into additional parts (rather than dropping notes) so no material is lost. This is one of the few places in the pipeline where the artefact schema is manipulated to keep a downstream tool happy; the manipulation is explicit, reversible on inspection, and documented in the merged score's `<part-list>`.

4.4 Round-trip identity F1 and score-vs-ground-truth F1

Two F1 numbers are reported for the merged score, and they measure different things:

- **Round-trip identity F1 (self-vs-self).** The 8-bar seed is round-tripped through xml → mid → xml → mid → xml twice. Note-level F1 comparing the final MIDI to the input MIDI is **1.00** for drums, bass, and other. This measures whether the notation and MIDI round-trips are lossless.
- **Score-vs-ground-truth F1 (transcription-bounded).** On the M-SEP-1 synth reference, note-level F1 comparing the merged score to the tiled ground truth is **upper-bounded by basic-pitch's per-axis F1 from Section 3.6**: pitch F1 ≤ 0.92 on drums, ≤ 0.86 on bass, ≤ 0.71 on other. The merged score does not recover notes basic-pitch did not emit; the notation stage is faithful to its input, not clairvoyant about its input.

Reporting these two numbers side by side is the honest form of the result: the notation-and-MIDI pipeline itself is exact, and the end-to-end score fidelity is bounded by the transcription stage above it.

4.5 The rules ledger: schema and identity

The rules ledger (M-RULES-1) is the pipeline's symbolic knowledge base. Each row is a typed rule spanning one of five categories:

- **harmonic** (chord progression, cadence, modulation),
- **rhythmic** (metre, hemiola, syncopation pattern),
- **melodic** (contour, interval, motif),
- **form** (repeat, section, arrangement layout),
- **arrangement** (instrumental role, tessitura, register).

Every rule row carries:

- a `rule_id` computed as the SHA-256 of the normalised rule body (so two independently-extracted identical rules deduplicate to one row),
- a `provenance` pointer to the merged-score fragment the rule was extracted from (song hash + measure range + part id),
- a `type` field naming one of the five categories,
- a `parameters` blob whose schema depends on the type,
- and an `emitted_by` field naming the extractor version.

The schema is validated at write time against a planted-invalid rejection matrix: for each of eleven synthetic malformations (wrong type, missing field, mistyped parameter, dangling provenance, non-canonical order, etc.), the writer must reject the row. This matrix is exercised on every schema change.

4.6 Extraction ledger growth: 28 → 76 rules

The first-generation extractor, run on the 8-bar seed alone, produced **28 rules**: 9 harmonic, 5 rhythmic, 7 melodic, 3 form, 4 arrangement. The second-generation extractor, run on three additional seeds drawn from the focus set, grew the ledger to **76 rules** with zero duplicate `rule_id`s and zero rejections against the planted-invalid matrix. The growth curve is sub-linear in seeds — as expected, since common cadence, metre, and motif rules recur across material — but does not plateau: the second-generation extractor's long tail includes several arrangement rules (drums-plus-bass unison downbeats, vocals-lead-then-doubled) that the 8-bar seed alone did not exhibit.

4.7 Concat hardening and the tiled generation path

Once the ledger reached 76 rules, a hardening step exercised the concatenation-and-projection path used by generation (Section 8): a sampled subset of rules is projected onto a tiled sequence of empty measures, the resulting merged score is round-tripped, and the identity F1 is checked against the pre-round-trip state. This closed a family of edge cases around tempo changes at tile boundaries and time signature restatement. Hardening is what took the merged-score bridge from "works on the seed" to "works on synthetically-generated inputs of arbitrary length".

4.8 What the reader should carry forward

- The merged-score bridge is byte-exact across two full round-trips and imposes no additional loss beyond transcription.
- The rules ledger is typed, provenance-tracked, and hash-deduplicated; 76 rules across five categories at the point of writing.
- Every downstream reasoning stage — rule-based part assignment in Section 5, texture-panel input generation in Section 6, palette choice in the DAW render — reads from the ledger, not from the merged score directly.

5. The DAW Stack and the Palette-Instrument Determinism Arc

5.1 The 2026-08-28 stack reversal

The project's original DAW choice was Ableton Live plus Pro Tools, driven by their strong plugin ecosystems and mature MIDI-clip authoring. That choice was reversed on 2026-08-28 in favour of **Ardour + DawDreamer + open-source plugins** (LV2 and open VST3s). Three reasons converged:

1. **Redistribution constraints.** Ableton and Pro Tools cannot be installed unattended in a reproducible workspace; every worker would have needed a manually-managed license seat. Ardour installs from package under an open licence, and DawDreamer is `pip install`-able.
2. **Determinism reach.** DawDreamer exposes plugin state as Python-native parameter maps that can be dumped, hashed, and diffed; commercial DAWs' automation formats

are opaque binary blobs whose byte-identity across saves is not guaranteed even without semantic change.

3. **Automation as first-class data.** DawDreamer's `set_automation` accepts a per-parameter time-series directly, so automation curves are inputs the pipeline can construct programmatically from the rules ledger, rather than clip-region drawings maintained by hand.

The reversal was carried out as a bounded validation spike — M-DAW-SPIKE-1 — rather than a wholesale rewrite; the spike's purpose was to answer "can this stack render a merged score with automation, deterministically, end to end" before any downstream code was re-pointed.

5.2 What the spike proved and its two documented gaps

The spike closed the end-to-end path for a single seed song: merged score → per-part MIDI → Ardour session with correct automation lanes → render to WAV → panel input. Two gaps surfaced during the spike and were documented rather than papered over:

- **GAP-1: Ardour Lua MIDI-file import.** Ardour's Lua scripting API does not expose a stable programmatic path for importing a standalone MIDI file into a new track region. Manual UI import works; scripted import does not (the API surface exists, but its behaviour across Ardour patch releases was not stable enough to build on).
- **GAP-2: Ardour VST3 automation delivery.** Even when a VST3 plugin is instantiated in an Ardour track and an automation lane is drawn onto it, parameter changes were observed not to reach the plugin's audio-thread state on render. Initial diagnosis pointed to VST3 specifically; later work sharpened this: the delivery gap affects **both VST3 and LV2** paths on the versions of Ardour available in the workspace.

GAP-1 was closed by a **redefinition**: rather than route MIDI through Ardour's importer at all, the pipeline pre-renders each MIDI part to audio via `fluidsynth` (with a pinned SoundFont per part) and then authors the Ardour audio-region XML by hand. Hand-authored XML uses Ardour's stable on-disk schema (which is stable across the patch releases we tested) rather than its unstable scripting API. The audit records this closure as a REDEFINED-GAP, not a fix — a signal to future readers that the underlying Lua import path is still not exercisable.

GAP-2 remains **STILL-GAP** at the time of writing. The pipeline works around it with a two-step render (rendered stems from fluidsynth, then DawDreamer for parameter-automation-driven effect passes on those stems); the workaround is documented in the DAW-stack README and is consumed by the palette-instrument work in §5.4.

5.3 DawDreamer `set_automation` gap-closure and `env_corr`

Independent of the Ardour VST3/LV2 gap, DawDreamer's own `set_automation` path was exercised as the primary automation-delivery mechanism. The gap-closure metric is `env_corr`: the Pearson correlation between the rendered stem's short-window RMS envelope and the target envelope encoded by the automation curve. Two bars were pre-registered:

- **Primary bar:** `env_corr ≥ 0.9` on a sinusoidal AM-envelope target.
- **Secondary bar:** `env_corr ∈ [0.15, 0.3]` — a floor that distinguishes "automation is delivered at all" from "automation has no measurable effect."

The observed number after gap-closure was **env_corr = 0.487**. This misses the primary bar and satisfies the secondary bar. The result is reported as a **partial closure**, not a pass: it demonstrates that `set_automation` reliably delivers a monotonic control signal to the plugin's audio thread, but the sinusoidal target's high-frequency detail is smoothed by parameter-quantisation and per-block latency in the plugins we tested. Downstream panel input is computed on the actual rendered envelope, not on the target — so the panel measurements remain faithful even where the primary bar is missed.

5.4 The palette-instrument determinism arc

Beyond the automation gap, the pipeline needed a palette of instruments — synths and samplers whose per-parameter state can be set programmatically and whose rendered output is byte-deterministic for a given parameter payload. Two plugins were adopted:

- **Surge XT** (open-source hybrid synth) for pitched tonal content.
- **Dexed** (open-source FM synth, DX7-family) for FM-specific timbres and for the leak-test control condition of a strong, spectrally-narrow reference tone.

Each is rendered via DawDreamer under a fixed sample rate (44.1 kHz), a fixed block size, and a fixed parameter payload. Byte-identity was verified across:

- **Independent invocations** on the same host (same worker, cold cache): identical bytes.
- **Salted invocations** across a set of five deterministic-salt seeds intended to perturb any hidden nondeterminism (thread scheduling, allocator interleavings, floating-point summation order): identical bytes.
- **Two hosts** with matching CPU family: identical bytes.

Both plugins passed. The determinism proof is conditioned on the CPU family being homogeneous across workers — the pipeline pins workers to a single family precisely because we did not want to promise cross-family byte identity we could not measure.

5.5 The c31 STILL_GAP-to-activation closure

The palette-instrument work started as a STILL_GAP: the fixture at c26 showed byte-identity on a synthetic sine-wave rendering, but the first attempt at rendering a real merged-score part through the same path exposed a mismatch between the fixture's parameter payload and the shape the real path fed the plugin (an additive-kwargs mismatch in `scripts/palette_render/render_stem.py`). The c31 activation closure resolved this: the render function was extended additively (new kwargs default to a value that recovers the fixture's behaviour), the fixture was re-run to confirm no regression, and a real merged-score part was rendered end-to-end with byte-identical output across the determinism grid above.

The audit notes one downstream consequence: the anchor manifest that pinned the pre-c31 form of `render_stem.py` was not republished after the c31 additive-kwargs edit. The manifest's contract still holds (`parameter_dict=None` recovers the pinned c33 anchor byte-for-byte), but this backwards-compat guarantee lives in the code and the audit, not in the manifest itself. Republishing the manifest as `_v2` with the post-edit SHA and an explicit backwards-compat clause is called out under future work.

5.6 A schema wart worth noting

The palette-instrument determinism milestone emits its per-song determinism verdict as a row in a TSV (`data/palette_determinism/ scorecard.tsv`) rather than as a top-level `verdict.json`. This is faithful to the milestone's rubric (which is defined per-row) but breaks the cross-milestone convention that verdicts live at the top level as JSON with a `rubric_hash` key. The audit flags this as verdict-schema drift; consolidating the TSV row into a wrapper `verdict.json` (referencing the TSV as an artefact rather than as the verdict itself) is a small future-work item that does not change the determinism guarantee.

5.7 What the reader should carry forward

- The DAW stack is Ardour + DawDreamer + open plugins, chosen for determinism reach and unattended reproducibility.
- One documented gap (GAP-1, Ardour Lua MIDI import) is closed by a hand-authored XML redefinition; one (GAP-2, Ardour VST3/LV2 automation delivery) remains open and is worked around by a two-step render.
- DawDreamer's `set_automation` delivers monotonic control signals (`env_corr = 0.487`, primary bar 0.9 missed, secondary bar 0.15/0.3 satisfied).
- Surge XT and Dexed render byte-deterministically across independent invocations, salt sweeps, and paired hosts within a pinned CPU family. This byte-identity is the load-bearing property that lets the generation stage in Section 8 hash renders and deduplicate at the payload level.

6. The Texture Panel (M-TEX-1)

6.1 The refuse-to-aggregate contract

The texture panel is the single largest concession this project makes to honesty over convenience. Every mature evaluation tradition — for speech synthesis, for image generation, for music tagging — offers a one-number scalar quality score. M-TEX-1 explicitly refuses that. The panel reports **a vector of side-by-side distances** on every comparison it makes, and it exposes the family disagreements between those distances rather than smoothing them into a mean.

The rationale is that no single mel-, spectral-, or embedding-derived distance measures texture faithfulness well enough to be trusted alone. Each family has known blind spots — mel L1 is insensitive to fine timbral colour, VGGish is content-dependent, CLAP over-weights semantic tags, the loudness envelope ignores spectral balance — and averaging their normalised values produces a number that looks authoritative and is not. The contract is therefore: **the panel is the answer; the answer is not a scalar**.

6.2 What the panel measures

The panel emits four families of distances per comparison, at multiple scales where the metric admits scale:

- **Multi-scale mel L1.** Log-mel spectrograms are computed at three window/hop pairs (2048/512, 4096/1024, 8192/2048), L1 distance is taken per scale, and all three are reported.
- **Spectral centroid, bandwidth, rolloff, flatness.** Four librosa-family low-order spectral statistics, each reported as frame-mean L1.
- **Loudness envelope.** Short-window RMS envelope L1, with an additional cross-correlation `env_corr` figure (the same statistic used to characterise the DawDreamer automation delivery in Section 5.3).
- **CLAP and VGGish perceptual embedding distances.** Cosine and L2 distances in each embedding space, reported separately (they disagree, and the disagreement is the point).

For a single pairwise comparison this produces on the order of a dozen scalar entries. The panel does not reduce them.

6.3 The stage-by-stage comparison

The comparison the panel is designed for is not “recreation vs original” alone. It is three-way: **original ↔ bare-MIDI ↔ effects-layered**, where:

- original is the source audio,
- bare-MIDI is the merged score of Section 4 rendered through the DAW stack of Section 5 with palette instruments but no effects chain,
- effects-layered is the same render with the rules-ledger-derived effects chain applied (EQ, compression, reverb, delay).

The three-way structure lets the panel disentangle two questions the project genuinely cared about separately: how much of the texture gap is symbolic (bare-MIDI vs original) vs how much is production (effects-layered vs bare-MIDI). On the seed song, roughly two-thirds of the mel L1 gap and roughly half of the VGGish cosine gap closed between bare-MIDI and effects-layered, consistent with production being a large but not dominant contributor to texture faithfulness.

6.4 Widening: three seeds, 72 finite numbers

Initial evaluation was on a single seed; widening to three seeds produced 72 finite scalar entries (three seeds × three stages × four families × the family-appropriate number of sub-scales). “Finite” is literal — the panel refuses to emit NaN and refuses to fall back to imputation. A metric that cannot be computed for a given comparison (for example, VGGish cosine on a stem below the model’s minimum duration) is reported as an explicit absent row, not as zero.

Across the three seeds the panel’s headline observation was that no single distance ordered the three stages consistently. Multi-scale mel L1 preferred effects-layered on all three seeds. VGGish cosine preferred effects-layered on two seeds and bare-MIDI on one. CLAP cosine preferred effects-layered on the two ensemble seeds and was essentially tied on the monophonic seed. This is the family disagreement the panel exists to surface.

6.5 The content-flip embedding analysis

The one persistent VGGish flip — bare-MIDI preferred over effects-layered on a specific seed — was drilled into as a dedicated sub-milestone: the content-flip embedding analysis. The finding is that the flip is **content-dependent, not seed-random**:

- On polyphonic material, VGGish’s cosine geometry places the effects-layered render closer to the original: the model rewards the spectral density that the effects chain adds.
- On monophonic decaying-triad material (the specific seed that flipped), VGGish places the bare-MIDI render closer: the effects chain’s reverb tail moves the embedding away from the dry-monophonic original along a dimension the model uses.

The analysis’s importance is not to fix the flip — it is a true fact about VGGish, not a bug in the pipeline — but to make it legible in the panel output. The relevant panel row now carries a `content_class ∈ \{polyphonic, monophonic-decaying-triad, mixed\}` tag that tells a downstream reader when this flip is expected.

One book-keeping note: the c14 clone-2 closure event pinned 13 artefacts plus an adopt event for a `variants/` directory of content-flip case studies. That directory is only partially present on disk today; the analysis’s findings survive intact in the verdict JSONs and in the tagged panel rows, but the illustrative per-case variants would need to be regenerated to restore the directory to its pinned form. The audit records this as evidence drift, tracked but not blocking.

6.6 A note on figures

The M-TEX-1 content-flip analysis has figures on disk under `docs/figures/`; the M-RECREATE-2 RC0..RC10 scorecards are tabular only and would benefit from before/after mel and centroid plots that we did not generate in-run. The audit's figure-coverage assessment flags this as an outstanding item; it does not change any numeric finding.

6.7 What the reader should carry forward

- The panel is the answer; there is no single texture number.
- Every panel row is a real measurement, not a smoothed average or an imputation.
- Family disagreements — especially VGGish vs mel L1 on monophonic decaying material — are surfaced by construction, not swept up.
- The panel is what Section 7's ear model is calibrated against as a separate signal from the ear model's own ordinal loss; keeping the two signals distinct is a load-bearing part of the ear-model design.

7. The Ear Model (M-EAR-1)

7.1 What M-EAR-1 is (and is not)

M-EAR-1 is an ordinal listener model: given a 30-second music chunk it emits a real-valued score whose rank — not its calibrated magnitude — is the object of interest. Trained on the operator's rated corpus (ear bands 6/5/4), it predicts whether one chunk is likely to be preferred over another by the same operator, and it is used downstream in Section 8 to rank generation candidates without requiring the operator in the loop for every sample.

Two disclaimers up front. First, M-EAR-1 is a listener model, not a music-quality model: it captures one specific operator's ordering, and the leak-test in §7.5 explicitly demonstrates it does not — must not — consume any of the seven non-factor fields from Section 2.4. Second, M-EAR-1 is deliberately held **in progress** at the report date, for reasons that are load-bearing and are the point of this section.

7.2 The CORN ordinal-regression head

The training objective is CORN: consistent rank logits for ordinal regression, in the sense of Cao/Mirjalili/Raschka. CORN decomposes an ordinal target of K classes into $K-1$ conditional binary classification tasks arranged as a staircase, and uses a single shared backbone with $K-1$ output logits whose monotonicity is enforced by construction. For this project's three-band operator rating (6/5/4), $K=3$ so the head has two output logits.

CORN is preferred over softmax-cross-entropy for two properties: the predicted class order is guaranteed monotone (a chunk cannot be scored "band 6 with probability 0.6 and band 4 with probability 0.7"), and the training loss is decomposable per boundary, so the two boundary decisions (band-6-vs-not, band-≥-5-vs-band-4) can be tuned independently by re-weighting per-boundary loss terms.

7.3 The Path B commit doc and the three-threshold pre-registration

At c26 the project committed to Path B: rather than tuning M-EAR-1's operating point post hoc, three thresholds SB1 / SB2 / SB3 were **pre-registered** in a commit document that pinned the exact `focus_set` and the exact test statistics under which the model would be judged:

- **SB1** — band-6 recall on the held-out slice.
- **SB2** — band-4 precision on the held-out slice.
- **SB3** — Spearman rank correlation between predicted scores and operator labels on the full available corpus.

Path B's discipline is that the three thresholds are set before observing any test-set metric. Once set, they are the model's acceptance criteria; the model is not tuned to satisfy them retrospectively.

7.4 EAR_v2_PARTIAL and the IMPROVEMENT criterion

M-EAR-1 v2 was evaluated against SB1/SB2/SB3 at c45. The verdict was **EAR_v2_PARTIAL**: v2 improves over v1 on all three thresholds simultaneously but does not clear the pre-registered PASS bar on any of them at full statistical confidence given the current corpus coverage.

A follow-up adjudication at c46 clarified the mapping: PARTIAL fires under an IMPROVEMENT criterion (v2 - v1 > 0 on each of the three metrics, tested independently), which is distinct from the PASS criterion (v2 exceeds the pre-registered SB1/SB2/SB3 thresholds absolute). The c47 sub-leaves (per-boundary calibration sub-milestones) individually validate on the smaller subsets they are scoped to. The parent M-EAR-1 is held **in progress by design** until real-label calibration on the full 80-song corpus can be run — which itself waits on the egress unblock discussed in Section 2.

The audit reflects this cleanly: M-EAR-1 in-progress / high confidence / 27 evidence rows; M-EAR-1/real-label-training-v2 in-progress / medium confidence / 6 evidence rows; three v2.1 sub-leaves validated at c47.

7.5 The leak test: non-factors go unread

The seven non-factor fields registered per song in Section 2.4 — genre, country, release date, language, instrumental-vs-lyrics, live-vs-studio, artist — exist chiefly so the ear model can be tested against them. The leak-test harness runs the following experiment on every candidate model:

1. **Plant a strong signal.** For each non-factor, construct an adversarial training/test split in which the field perfectly predicts the operator rating on the training half (rating and field are correlated at $r = 1$) and is decorrelated on the test half.
2. **Run the model unchanged.** Train the ear model on the training half with the non-factor fields present in the sidecar (as they always are), evaluate on the test half.
3. **Require failure to detect.** If the model has consumed the planted non-factor even indirectly, its test-half accuracy will collapse (the correlation the model latched onto is not there). The leak-test requires that test-half accuracy remain indistinguishable from the no-plant baseline. Passing the leak test means the model did not find the signal it was invited to.

M-EAR-1 v2 passes the leak test on all seven non-factors at the confidence level set by the training-half plant strength. This is reported as a positive result about the model's inductive architecture, not about any specific corpus split: the leak test manipulates the split precisely to make the plant strength independent of natural correlations in the corpus.

7.6 The armed-harness sub-milestone

M-EAR-1/armed-harness is the mechanical readiness state that will enable a full real-label calibration run automatically once egress opens. Its readiness gate is precisely the two-consecutive `media_ok=true` production probes described in Section 2.5. It is fixture-verified at c26 and re-verified at c31 (the fixture emits a synthetic ratings-manifest CSV and a synthetic 80-song audio bundle, runs the calibration end-to-end, and checks that the resulting model lands its verdict under the pre-registered rubric hash). The sub-milestone is in-progress because its production-mode gate has not fired.

7.7 The corpus gap and its downstream consequences

At the report date 43 of the 80 rated songs have on-disk audio; the remaining 37 have full provenance rows in `corpus/ratings/ratings_manifest.tsv` but blocked audio. The corpus gap directly limits three things:

- **Statistical power of SB1/SB2/SB3.** The pre-registered thresholds were chosen with an 80-song evaluation in mind. On 43 songs the confidence intervals around each threshold's test statistic are wider by roughly a factor of $\sqrt{80/43} \approx 1.36$, and the PASS bar is therefore stricter in effect than in intent.
- **Band-boundary calibration.** The band-4 slice is the smallest of the three in the operator's distribution; losing any of its songs to egress makes the band-4 boundary the tightest to calibrate cleanly. The c47 sub-leaves that individually validate at high confidence are, in effect, the band-6/band-5 boundary work, precisely because it has the most surviving data.
- **Full-corpus real-label training.** The v2 head was trained on the 43 available songs; a v3 head on the full 80 songs is the gated next step. The commit-doc rubric hash for v3 is registered and the ARMED harness is ready; only the egress gate remains.

The audit records a separate provenance drift here: the manifest is missing 10 band-7 rows despite RECEIPTS.md claiming an update. Band 7 is the exemplar-only ceiling from Section 2.1 and is used as a positive-control channel for M-RECREATE-1/first-real-audio rather than as a training band for M-EAR-1, but the reconciliation belongs in future work either way.

7.8 What the reader should carry forward

- M-EAR-1 is an ordinal listener model with a CORN head and pre-registered PASS bars SB1/SB2/SB3.
- The current verdict is EAR_v2_PARTIAL: v2 monotonically improves on v1 across all three bars but does not clear PASS at full confidence given the corpus coverage.
- Non-factor leak testing is a load-bearing part of the design and is passing.
- The parent milestone is held in-progress by design, pending an automatic full-corpus real-label calibration whose only remaining gate is the egress-ready state machine of Section 2.5.

8. Recreation, the Accurate-Small-Set Programme, Generation, and the Collision Arc

8.1 The recreation-before-generation bet

The project's structural bet — set at the top of this report — is that faithful recreation of an existing rated song is a prerequisite for meaningful generation. Sections 3 through 7 built the toolchain; Section 8 exercises it end-to-end. Two milestone families do the exercising: M-RECREATE-1 recreates single rated songs to prove the pipeline closes on real audio at all, and M-RECREATE-2 (accurate-small-set) hardens the closure on a five-song focus set with pre-registered per-stage rubrics.

8.2 M-RECREATE-1: first end-to-end recreation

The first end-to-end recreation of a rated song ran on the band-7 exemplar `016__LOCAL__05_02.mp3` and closed with verdict **RECREATION_LANDS**. The headline number is Δmel_l1_db $= +5.906$ dB improvement of the effects-layered render over the bare-MIDI render (measured against the original). Both terms are texture-panel measurements (Section 6); the improvement is exactly the kind of "how much does production add" question the three-way stage-by-stage comparison was built to answer, now on a real rated song rather than the 8-bar seed.

The audit records one downstream schema drift for this milestone: `verdict.json` lacks a top-level `rubric_hash` key, breaking the three-way byte-equality convention used by later milestones. The verdict content is intact and reproducible; the schema wart is tracked as future work, not as a substantive result to revisit.

8.3 The M-RECREATE-2 accurate-small-set programme

M-RECREATE-2 hardens recreation on a five-song focus set drawn from bands 6 and 5. The programme is decomposed into recreation cells RC0–RC10, each with a pre-registered rubric and a per-song scorecard:

- **RC0** — baseline: current-pipeline recreation without per-stage hardening. Sets the floor every RCn has to beat.
- **RC1** — vocals transcription (Branch A, verdict RC1_RC9_LANDS, **4/5 focus songs**).
- **RC2** — drums stem transcription.
- **RC3** — bass stem transcription.
- **RC4** — GM program map (folded into RC1/RC2/RC3 substantive branches; the c50 stub records the intent).
- **RC5** — tempo/beat grid estimation (Branch B carried the substantive tempo work at c51).
- **RC6** — panel-gate: verifies that Section 6's panel does not regress on the RC1–RC3 outputs.
- **RC7** — mix balance (EQ + loudness match): **5/5 focus songs, 20/20 stem accepts** on substantive per-stem MIDIs, rerun at c1-1_clone_0 and reconfirmed.
- **RC9** — first-class parts (per-part identity and role tagging): **5/5 focus songs** under Branch A of c51.
- **RC10** — transcription real-stem resurvey with post-processing: drums-and-bass impl-per-stem, winner-per-stem selection, and post-processing applied. Validated end-to-end at cycle 54 with five sub-milestones landing under the accurate-small-set parent (`drums-bass-pre-registration`, `-impl-per-stem`, `-winner-selected`, `-post-processing-applied`, `-verdict-emitted`).

Two book-keeping notes carried by the audit and named here for completeness: (a) the plan pre-registered an accurate-small-set-v2 supersede parent that never emitted its own firing events (all c50+ leaves fired under the v1 parent), a plan-ledger drift slated for a single supersede event in future work; and (b) the RC0..RC10 scorecards are tabular only — before/after mel and centroid plots would help a reader as figures and were not generated in-run.

8.4 M-GEN-1: the generation-batch arc

M-GEN-1 exercises generation independent of recreation quality: it draws from the rules ledger (Section 4), constructs candidate merged scores, renders them through the DAW palette (Section 5), and scores the results with the ear model (Section 7) and the texture panel (Section 6). Six batch revisions were produced:

- **batch-v1** — baseline: random rule draws, uniform palette.
- **batch-v2** — rule-cluster-conditioned draws (per rule-type clusters extracted from the ledger).
- **batch-v3** — first attempt at palette-conditioned rendering (palette chosen by rule-cluster).
- **batch-v4** — palette-driven-v4: palette chosen by rule-cluster and re-conditioned on the M-EAR-1 v2 ranking of candidate palettes on prior batches.
- **batch-v5** — batch-vs-cluster: measured whether cluster-anchored draws Pareto-dominate uniform draws on the panel; result was mixed by content class, consistent with the Section 6.5 content-flip finding.
- **batch-v6** — the current top: palette-driven, cluster-anchored, ear-ranked, with a deduplication pass on payload hash.

Every batch's verdict JSON and provenance row survives on disk. The audit notes that **684 ledger-referenced generation artefacts are absent from disk** (artifact loss under M-GEN-1/batch-v-cluster): these are the rendered audio outputs themselves, which are deterministically regenerable from the seeded ledger by re-running the render sweep. The verdicts and the panel scores that gate the verdicts are intact; only the audio bytes downstream of them would need to be re-materialised. A single deterministic sweep re-produces them, and the regeneration is called out as a discrete future-work item.

8.5 The collision-modelling arc: PARTIAL_BP_UNRESOLVED_SHAPE

Independent of both recreation and generation, one investigation ran its course to a fully-negative published outcome and deserves naming: the collision-modelling arc. The question was whether observed generation-output collisions (two different rule-payload draws producing byte-identical rendered audio) are explained by any of four structural mechanisms:

- **M1 — coherence-gate coercion.** Hypothesis: the coherence gate in the palette-choice step deterministically maps distinct payloads to the same rendered output. Verdict: **structurally disqualified** — the gate's output alphabet is strictly larger than the number of observed collisions per bucket, so it cannot be the exhaustive cause.
- **M2 — effective-K.** Hypothesis: the effective number of distinct render outputs K_{eff} per rule-type is small enough that a birthday-paradox collision rate matches observations. Verdict: **refuted** — measured K_{eff} per rule-type is at least an order of magnitude larger than what would reproduce the observed collision rate.
- **M3 — hash-space geometry.** Hypothesis: colliding payloads cluster geometrically in the hash space, per (rule_type × salt) cell. Verdict: **collapsed under multiple-testing correction** — apparent per-cell clustering did not survive correction across the 40+ cells tested.
- **M4 — semantic-cluster overlap.** Hypothesis: colliding payloads share a semantic cluster in the rules ledger's cluster space (Section 4). Verdict: **refuted** — collision rates within and across clusters are statistically indistinguishable.

The final verdict on the arc is **PARTIAL_BP_UNRESOLVED_SHAPE**: the observed collision rate is real, but no candidate mechanism accounts for it, and the arc is closed with the negative result published rather than left to accumulate follow-ons. This is the honest form of the finding: a genuinely-open shape question about the distribution of collisions, with the four best-motivated candidates each independently ruled out. The candidate list is what was ruled out, not a full search of the hypothesis space; the arc's closure document is explicit that unexamined mechanisms remain possible.

8.6 What the reader should carry forward

- The pipeline closes end-to-end on a real rated song (M-RECREATE-1, +5.906 dB effects-over-bare on the band-7 exemplar).
- The five-song focus set is hardened across RC0–RC10 with per-stem pre-registration; RC7 (mix balance) and RC9 (first-class parts) both land 5/5 focus songs.
- Six generation batch revisions were produced; verdicts and provenance are intact and the 684 missing audio artefacts are deterministically regenerable from the seeded ledger.
- The collision-modelling arc closes with a published negative result across four structurally-motivated candidate mechanisms. The unexplained-collision distribution remains a live open question, carried into future work.

9. The v4 Closure Campaign

9.1 Framing: what a "closure" campaign is for

The pipeline described in §§2–8 is the v3 system: an end-to-end path from an ingested reference recording to a rendered re-creation with a documented ear, a validated texture panel, and an accurate small-set generation arc. By the end of that work the operator had heard, and accepted, the v3 Chicken Grease reconstruction. What remained was not another pass at the pipeline itself but a bounded set of follow-on deliverables that use it: per-instrument sound matching against pinned render families, one full-song A/B showcase mix, a rules artefact expressed in the v4 sound-matching vocabulary, a lightweight exemplar ear, and a small seeded-generation batch. The v4 closure campaign is that follow-on set. Its remit was explicitly bounded — deliver these items, then end the run cleanly — and this section reports the state each deliverable reached before that termination.

The organising unit of the closure campaign is the **per-instrument arc**: for a given song and instrument, search two frozen render families for a configuration whose short (6 s) rendered clip best matches the reference stem under a fixed composite objective, then adjudicate the result against a frozen decision protocol. The two families are (1) a General-MIDI SoundFont sweep over the standard bank of GM programs — the "sf2" family — and (2) a stem-sampled concatenative builder that constructs a slice bank from the reference stem's own onsets and dispatches each MIDI event to the nearest-pitch slice with pitch-shifting — the "family-2" family. The decision protocol has three terminal outcomes:

- `CONFIRMED` — the best candidate's VGGish embedding-cosine score against the reference stem is at or above 0.60. The pinned configuration becomes the delivery audio for that instrument cell.
- `RULED_OUT` — the best candidate is at or below 0.40. The family is dropped from consideration.
- `STILL_INDETERMINATE` — the score falls between the two bars. No commitment either way.

When both families are `RULED_OUT`, the arc closes as `EXHAUSTED_NO_CONFIRMED` and a pre-registered options fork is opened. One option in every such fork is a **refuse and substitute** ruling: decline to synthesise the instrument and splice the operator-heard reference stem verbatim into the showcase mix. Refuse-and-substitute is not a fallback — it is a first-class outcome that preserves the operator's ear as the authoritative reference for that instrument.

Every configuration selected under this policy is pinned as a **deterministic replay proof**: two independent renders of the same profile under a canonical seven-key environment pin (`LC_ALL`, `MKL_NUM_THREADS`, `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, `PYTHONHASHSEED`, `SOURCE_DATE_EPOCH`, `TZ`) must produce byte-identical output WAVs. The environment pin has hash prefix `2ac444c36298d6ad…` and has been in force since early in the campaign.

9.2 The determinism certificate

The first closure deliverable was a bit-exact determinism certificate for the v3 Chicken Grease reconstruction, obtained by running the full delivery pipeline twice with all caches disabled and hashing the output WAVs. Both renders produced byte-identical output. The certificate is complete and confirmed as of the campaign's opening.

The certificate matters because it is the invariant every subsequent v4 deliverable rests on: if a pinned profile's replay proof does not reproduce, the failure is in the profile or the pin, not in the pipeline underneath.

9.3 The Chicken Grease per-instrument arcs

Chicken Grease was the sole song carried all the way through per-instrument arc closure during the campaign. Its six `htdemucs` stems (bass, drums, guitar, piano, other-residual, vocals) were each taken to a terminal verdict; the outcomes, in the order they were closed, are the following.

Bass — accepted (hybrid). The bass arc closed early under an explicit operator directive: accept the pinned SoundFont configuration (GM program 33) as the delivery bass, with the operator retiring the standard acceptance threshold for this specific arc and this specific instrument. This is the sole place in the campaign where the composite-objective ranking was used to select delivery audio rather than being used to either confirm or rule out a family; the directive's scope is narrow (CG-bass only) and does not extend to any other arc.

Drums — refused and substituted. The SoundFont sweep found a best candidate (GM program 16, "Power Kit") with an embedding-cosine of **0.2374** against the reference drums stem; the stem-sampled family-2 builder found a best rendered match at embedding-cosine **0.0372**. Both sit well below the 0.40 floor, and the arc closed as `CG_DRUMS_ARC_EXHAUSTED_NO_CONFIRMED`. The pre-registered acceptance fork was resolved by substituting the operator-heard `htdemucs` drums stem verbatim into the showcase mix.

Guitar — refused and substituted. The SoundFont fine-fit placed a muted-electric variant (GM program 28) as the top-1 configuration at embedding-cosine **0.2584**; the stem-sampled family-2 render scored embedding-cosine **0.0354**. Again both fell below the floor; the arc closed as `CG_GUITAR_ARC_EXHAUSTED_NO_CONFIRMED` and was resolved by refuse-and-substitute.

Piano and other-residual — grounded null. Both reference stems tested inaudible: the `pyloudnorm` LUFS-I measurement returned non-finite ("silence-only" content), and the RMS-dBFS fallback measured piano at **−81.53 dBFS** and other-residual at **−81.73 dBFS**, both far below the −60 dBFS silence floor. The corresponding v3-transcribed MIDI tracks carry zero note-on events. With no audible reference and no MIDI target, no sweep is warranted, and the delivery uses the (silent) `htdemucs` stem verbatim — the same treatment the v3 pipeline already applies to empty tracks.

Vocals — hybrid overlay (pre-existing). Vocals were covered by a policy established earlier in the pipeline: the `htdemucs` vocal stem is overlaid on the instrumental mix verbatim, without a synthesis attempt.

The net Chicken Grease showcase composition is therefore two synthesised cells (bass and — via the drums-substitution policy — the operator-heard drums stem re-used as-is), two grounded null cells (piano, other-residual), one substituted cell (guitar), and one overlaid cell (vocals). The delivery script's smoke test reports every cell terminal.

The pinned-profile schema. Supporting the arc-closure discipline is a pinned-profile JSON schema (v1) and its validator. The schema is deliberately permissive: it validates shape without pinning threshold semantics, so that it does not need to be revised when the metric-semantics question of §9.5 resolves. Every pinned profile from Chicken Grease's closure — bass, drums, guitar, and the two null pins — passes the validator.

Agent-picks selection invariants. Because the closure campaign was run under a hard rule against pausing for operator input on questions the operator had not been asked, the acceptance-fork resolutions above were made by the agent under a small set of codified invariants:

- (a) **No operator-scope extension.** A worker does not widen the scope of a directive the operator issued at a narrower scope.
- (b) **Prefer above-floor.** When one option selects a candidate below the retained absolute floor and another selects an above-floor candidate or takes a non-candidate policy path (such as refuse-and-substitute), prefer the latter.
- (c) **No misread rejection.** Do not reject an option based on a paraphrase of its own pre-registered text.
- (d) **Disclose on-disk-vs-brief divergence.** When on-disk state and a working brief disagree, disclose the divergence and pin the on-disk value by hash rather than silently converging.
- (e) **Additive-only extension of permissive schemas.** Extend a permissive schema by adding fields, not by tightening enforcement in place.

Invariants (a)–(c) were codified in response to an initial misresolution of the drums fork and validated on the very next fork (guitar), which resolved to refuse-and-substitute on the first attempt. Invariants (d) and (e) were added as comparable divergences appeared in later cycles. The invariants sit under, never above, operator authority.

9.4 The Chicken Grease A/B full-song showcase

The showcase deliverable — a stereo A/B mix that a listener can play against the original recording — was rendered from the closed per-instrument cells above. The delivered artefacts, all recorded as permanent read-only anchors, are:

- **`cg_ab_mix.wav`** — the mix itself, SHA prefix `6e13e0075c5d8116…`, located at `data/v4/deliveries/31a164f845f8e27e` `ab_mix.wav`.
- **`cg_ab_mix.manifest.json`** — records the inputs (bass configuration, drums-substitute stem, guitar-substitute stem, piano null, other-residual null, vocals overlay) that produced the WAV.
- **`cg_ab_mix.replay_proof.json`** — a proof that a fresh render from the recorded inputs reproduces the delivered WAV byte-for-byte.

Byte-determinism was subsequently re-verified a second time by a from-fresh-subprocess full-render regression suite covering the WAV, the manifest, the replay proof, the bass-gain amplification constant (`amplif = 2.688385`), the three pinned profiles that feed the delivery, and the discipline guards (no PRNG, absolute-path interpreter guard, canonical seven-key environment pin). A LUFS-I diagnostic sidecar measured the full mix at **−15.32 LUFS-I** without mutating the WAV; the per-stem loudness measurements are consistent with the mix, including the audibility-grounded nulls for piano and other-residual.

Every internal gate for the showcase — replay-proof byte-identity, all required inputs present, all discipline scans clean, all tests green — is satisfied. The remaining `LANDS` trigger is stated policy: the closure of a listening deliverable requires the operator's ear on the WAV. That trigger has not yet fired. This is a policy handoff, not an outstanding engineering task, and no substitute for it exists inside the system.

9.5 The four remaining focus songs and the metric-semantics escalation

The campaign's remit called for per-instrument arcs on four further focus songs — WIG, Rome, Disco A, and Peach Dream — in addition to Chicken Grease. Each of the four has been opened at **skeleton stage**: an addressable per-song directory under `data/v4/profiles/<song-hash>/` with a `stem_manifest.json` listing the six `htdemucs` stem hashes. No stage-1 sweep has been launched on any of them, and no thresholds have been committed for any of them. The skeletons are gated on a candid correctness concern about the composite objective that arose during the guitar arc and that the agent declined to resolve unilaterally.

The concern. The composite objective computes an embedding term from a pretrained VGGish audio encoder. The underlying panel implementation computes

$$\texttt{embedding\_cos\_vggish} \;=\; 1 - \cos(u, v)$$

which is a **distance** in $[0, 2]$ (lower is more similar, zero is identical). The composite objective consumes it correctly as a distance: it enters with a positive weight and the objective is minimised. However, the frozen decision protocol — the same one used in §9.1 to define `CONFIRMED` and `RULED_OUT` — expresses its thresholds (`≥ 0.60 CONFIRMED`, `≤ 0.40 RULED_OUT`) in the vocabulary of a similarity (higher is better). If the field is truly a distance, the guitar family-2 value of 0.0354 corresponds to a cosine similarity of ≈ 0.965 — extremely close to the reference — and the ruling should read `CONFIRMED`, not `RULED_OUT`. Under the current application, the closest candidates are being ruled out precisely because they are close.

Why this was not fixed inline. The correction is not a one-line change the agent could quietly apply, for two reasons. First, the closure campaign has a binding rule against retuning frozen numeric thresholds without cause. Second, and more consequentially, the correction has two different shapes and they imply different follow-through:

- **Path A — keep the field as distance and invert the threshold consumers.** Rewrite every downstream verdict- emitting site so that it applies the thresholds as distance bars (`≤ 0.40 CONFIRMED`, `≥ 0.60 RULED_OUT`) and re-adjudicates each Chicken Grease family verdict on the inverted floors.
- **Path B — apply a `1 - distance` correction at the emission point and leave the thresholds as written.** Change the panel or its immediate consumer to emit similarity rather than distance, re-issue the deterministic replay proof for the composite objective (because the numeric contract of every future pinned profile changes), and re-adjudicate every prior Chicken Grease family verdict on the intended similarity scale.

Both paths are internally consistent. They differ in what they imply about the historical verdicts, in whether the determinism certificate must be re-issued, and in what a future profile's numeric contract looks like. Choosing between them is an authority decision, not an engineering one, and it was recorded as an operator-authority escalation.

Why the showcase is safe under either path. The Chicken Grease showcase does not depend on the choice. The two `RULED_OUT` arcs (drums, guitar) were resolved by refuse-and-substitute, and the audio delivered for those cells is the operator-heard `htdemucs` stem verbatim — the authoritative reference under any interpretation of the composite metric. The bass arc was accepted under an operator directive that used the composite ranking but did not depend on the numeric floor. The two null cells are grounded in `pyloudnorm` measurements below the silence floor, not in the embedding metric at all. The vocals cell is under a pre-existing hybrid-overlay policy that does not consult the composite objective. The showcase therefore stands on any metric-semantics resolution.

What is blocked. The four remaining focus songs are what the metric-semantics decision blocks in practice. Their stage-1 sweeps have not been launched because a sweep run under one path would need to be discarded under the other; a sweep run under both paths in parallel would double the audio-storage and compute cost of the campaign's most expensive stage. Each skeleton records the blocker explicitly in a `blocked_on` field so that no downstream stage-1 sweep can proceed until the escalation resolves.

9.6 The v4 rules layer

The v4 rules artefact — a machine-readable extract of the generative rules the campaign's pinned profiles imply — was scaffolded at an early cycle and passed a scaffold-level smoke test. A substantive extraction pass followed on disk, producing extractor code and output artefacts that go materially beyond the scaffold. The substantive pass, however, has not been formally registered against the rules milestone's success criteria: there is no verdict row against the substantive extractor, no register-row for its output artefacts, and no supersede event linking the substantive extractor back to the scaffold it replaces.

This is a bookkeeping gap, not a correctness gap. The substantive extractor is present in the repository and its outputs are on disk. What is missing is the audit-trail formalisation that would let a future reader see, at the milestone level, that the scaffold has been replaced. The correct remediation — emit the missing verdict, register-row, and supersede event, and record an anchor-preservation snapshot confirming that the scaffold's smoke-test hash is unchanged — is a one-cycle housekeeping task and is enumerated in the future-work list of §10.

Separately from the substantive extractor, the pinned-profile schema described in §9.3 (`pinned_profile_schema_v1.json` and its validator) is a validated, load-bearing part of the v4 rules layer: it is what lets a downstream reader parse a pinned profile from disk and check it for structural compliance without re-running the pipeline that produced it.

9.7 Not started: exemplar ear, seeded generator, campaign closure

Three of the closure campaign's milestones were not begun.

Lightweight exemplar ear. A scoped ear model, distinct from the CORN ordinal head described in §7, built as a one-hour-compute exemplar over the five focus songs (Chicken Grease, Molasses, Essence, Desire, Peach Dream) using a CLAP + VGGish embedding ensemble with top-k window similarity, evaluated by a leave-one-out protocol requiring at least four of five held-out songs to score at least 6 and none to score below 5.5. This milestone does not depend on the metric-semantics escalation. It was scheduled after per-instrument profiles completion and remained unopened when the run ended.

Seeded generator. A generator program seeded from the v4 rules artefact, tasked with producing five novel instrumental songs and one interpolation-hybrid demonstration, with a stall rule after eight iterations without five passing candidates. It depends on both a substantive v4 rules verdict (§9.6) and the exemplar ear.

Campaign closure roll-up. The final closure deliverable — a completion report, an index of the delivered artefacts, the certificate status, and a list of the remaining gaps — together with updates to the operator-decisions document and the codebase guide, a final housekeeping sweep, and clean termination of the run. It depends on every v4 predecessor above.

Each of these three has a designed acceptance shape recorded in the campaign's plan of record, so a future resumption is a matter of execution rather than re-planning.

10. Conclusions, Honest Limits, and Future Work

10.1 What is done

The end-state has two layers: the v3 pipeline delivered against its "what counts as done" criteria, and the v4 closure campaign that sits on top of it and pursues bounded follow-on deliverables.

v3 pipeline (unchanged from baseline). Against the seven project criteria the run's v3 end-state is:

1. **Ingestion, classification, and provenance chassis exist and are honest.** Every chunk is content-addressed; the three-model classifier ensemble gates release; the seven-non-factor sidecar is stored and audited but never consumed. The egress-ready state machine is deployed and fires under a two-consecutive-`media_ok` rule.
2. **Source separation is deterministic and licensed for redistribution.** `htdemucs_6s` renders four stems byte-identically across independent invocations. Determinism was verified on a five-song slice; the weights fetch is reproducible.
3. **Transcription has an honest per-axis F1 on the M-SEP-1 synth reference.** basic-pitch 0.4.0 delivers usable pitch/onset/offset under a tuned octave-suppression grid; timbre, dynamics, and form are named as under-covered.
4. **A merged-score bridge is byte-identical across two full round-trips**, and a typed rules ledger of 76 hash-deduplicated rules is validated against a planted-invalid rejection matrix.
5. **The DAW stack renders deterministically on Surge XT and Dexed through DawDreamer**, with one closed gap (GAP-1 Ardour Lua MIDI import, closed by hand-authored XML) and one open gap (GAP-2 LV2/VST3 automation delivery, worked around by two-step render).
6. **The M-TEX-1 panel refuses to aggregate**, reports 72 finite panel entries across three seeds, and surfaces the VGGish content-flip as a labelled, understood family disagreement.
7. **The pipeline closes end-to-end on a real rated song** (M-RECREATE-1, +5.906 dB effects-over-bare on the band-7 exemplar) and the five-song accurate-small-set programme is hardened per-stem with RC7 and RC9 both landing 5/5.

v4 closure campaign. On top of the v3 pipeline the closure campaign completed the determinism certificate (§9.2), closed every per-instrument arc on Chicken Grease with a mix of acceptances, refuse-and-substitute rulings, and grounded nulls (§9.3), and delivered the Chicken Grease A/B full-song showcase on internal gates pending an operator ear (§9.4). It opened skeleton stem manifests for the four remaining focus songs but did not launch any sweeps against them, because a correctness question about the composite objective's metric was surfaced and correctly deferred to operator authority (§9.5). The pinned-profile schema is validated and load-bearing for replay discipline; a substantive v4 rules extraction is on disk but its verdict against the rules milestone's success criteria has not been formally emitted (§9.6). The exemplar ear, seeded generator, and campaign closure roll-up milestones were not begun (§9.7).

10.2 The three live constraints

Three constraints are load-bearing and honest.

- **Real-label M-EAR-1 calibration depends on the full 80-song corpus.** 43 of 80 songs have on-disk audio; the remaining 37 are registered with full provenance but their audio is behind the workspace egress policy. The armed harness (§7.6) will fire the full calibration automatically once the two-consecutive `media_ok=true` production probes land. M-EAR-1 is held in-progress by design until that fires.
- **Generation quality depends on the recreation loop closing on held-out songs.** M-RECREATE-1 closed on one band-7 exemplar and the five-song focus set is hardened; the accurate-small-set parent remains in-progress under the peer-under-G1 convention until the panel-gate cell (RC6) validates on the held-out songs under RC1..RC3 outputs.
- **The v4 metric-semantics escalation blocks the four remaining focus-song arcs.** The composite objective's `embedding_cos_vggish` field is computed as a distance but consumed by downstream decision protocols as a similarity. The remediation is an authority choice between two internally-consistent paths (§9.5) with different consequences for prior verdicts and for the determinism certificate. Until the choice is made, launching sweeps on WIG, Rome, Disco A, or Peach Dream would risk producing work that must be discarded under one of the two paths.

10.3 Actionable next steps

Drawn directly from the final audit's `future_work` block for the v4 closure campaign, in the order the auditor recommends attempting them, with the v3 follow-on items preserved underneath.

v4 closure campaign — in order.

1. **Operator selects between Path A and Path B for the metric-semantics escalation.** Path A keeps `embedding_cos_vggish` as a distance and rewrites every threshold consumer to apply the bars in the inverted sense (`≤ 0.40 CONFIRMED`, `≥ 0.60 RULED_OUT`), then re-adjudicates every prior Chicken Grease family verdict on the inverted floors. Path B applies a `1 − distance` correction at one emission point (`objective.py` or `embedding_panel.py`), re-issues the determinism certificate, and re-adjudicates every prior Chicken Grease family verdict on the intended similarity scale. Prior refuse-and-substitute pins for drums and guitar are safe under either path.
2. **Emit the missing substantive v4-rules audit-trail rows.** The substantive extractor is on disk; add the missing milestone verdict row against `M-V4-RULES-1/substantive`, a register row for its six output artefacts, a supersede event pointing back to the scaffold, and an anchor-preservation snapshot confirming the scaffold's smoke-test hash is unchanged.
3. **Run stage-1 and stage-2 sweeps on WIG, Rome, Disco A, and Peach Dream** once (1) resolves, under the same sweep-storage hygiene protocol that held during the Chicken Grease closure (score-and-delete; ≤ 500 MB working audio at any one time; a `df` check before each stage; working volume held under 90% full).
4. **Build the lightweight exemplar ear** over Chicken Grease, Molasses, Essence, Desire, and Peach Dream, using a CLAP + VGGish ensemble with top-k window similarity; no corpus calibration; leave-one-out acceptance is "at least four of five held-out songs at

score ≥ 6, none < 5.5."

5. **Build the seeded generator program** from the v4 rules artefact; deliver five novel instrumental songs and one interpolation-hybrid demonstration; stall the run after eight iterations without five passers.
6. **Execute the campaign closure roll-up:** completion report, indexed deliverables, certificate status, remaining gaps; refresh the operator-decisions document and the codebase guide; final housekeeping sweep; end the run.
7. **One housekeeping cycle for POR transcription drift** (five rows): canonicalise the v3-rules spec path in the plan-of-record narrative; refresh the five pinned SHAs against the on-disk artefacts; correct a malformed 62-hex drums-MIDI hash to its 64-hex on-disk value.

v3 pipeline follow-on items — preserved.

8. **Real-label M-EAR-1 calibration on the full 80-song corpus** per the c26 Path B commit doc — awaits egress unblock or manual manifest reconciliation.
9. **Add `probe_kind ∈ {smoke, production}` to `data/ingestion/egress_status.jsonl`**, so the two-consecutive-`media_ok` unblock signal cannot be spuriously satisfied by smoke rows.
10. **Rebuild the missing `data/gen/*` renders on demand from the seeded ledger** — a single deterministic sweep re-materialises them.
11. **Emit a single supersede event** that either renames the c51+ RC7/RC10 leaves to the pre-registered `accurate-small-set-v2` parent, or explicitly folds v2 back into v1 with a note that rubric-v2 was carried inline under v1 leaf identifiers.
12. **Restore or supersede the missing SSoT writer sources** (`long_exposure/workspace_bootstrap.py`, `long_exposure/tools/_ledger_schema.py`); if they were consolidated into surviving package modules, emit a `_plan/*-supersede` event that names the current SSoT.
13. **Republish `data/anchor_manifest_v1.json` as `_v2`** with anchor #20 = post-c36-edit SHA of `scripts/palette_render/render_stem.py`, and encode the backwards-compat contract (`parameter_dict=None` ≡ c33 anchor) explicitly.
14. **Append 10 band-7 rows to `corpus/ratings/ratings_manifest.tsv`** so provenance matches the on-disk audio M-RECREATE-1 consumed.
15. **Emit a closure event** adjudicating the two observed silent-death cases under `_manager/background-job-supervision-clone-0` (c31 fixture, c36 feature extraction), or archive them with lessons learned.
16. **Publish SSoT schemas for `anchor_preservation_v1.json` and `verdict_v1.json`** and have subsequent cycles conform.
17. **Fill the c41/c42 reporting gap, add the c52 egress-probe row, and either produce substantive c55-c58 content or retire the empty `report_cycles_56-58.md`.**

10.4 No new hypotheses

Everything above is drawn from what was measured. The one live scientific open end — the collision-arc PARTIAL_BP_UNRESOLVED_SHAPE (§8.5) — is left open with its four ruled-out candidate mechanisms named, and no new mechanism is proposed here. It is a real question about the distribution of generation-output collisions, and it deserves an unhurried follow-up rather than a speculative closure inside this report. The v4 metric-semantics question (§9.5) is likewise left as a named, adjudicable choice rather than resolved from inside the report; the two paths are stated in enough detail that the adjudication is a decision, not a research task.

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Internal artifacts and prior reports are referenced inline by their milestone identifier (`M-*`), stage cell (`RC*`), or cycle-report filename (`report_cycles_*.md` under `reports/cycles/`).